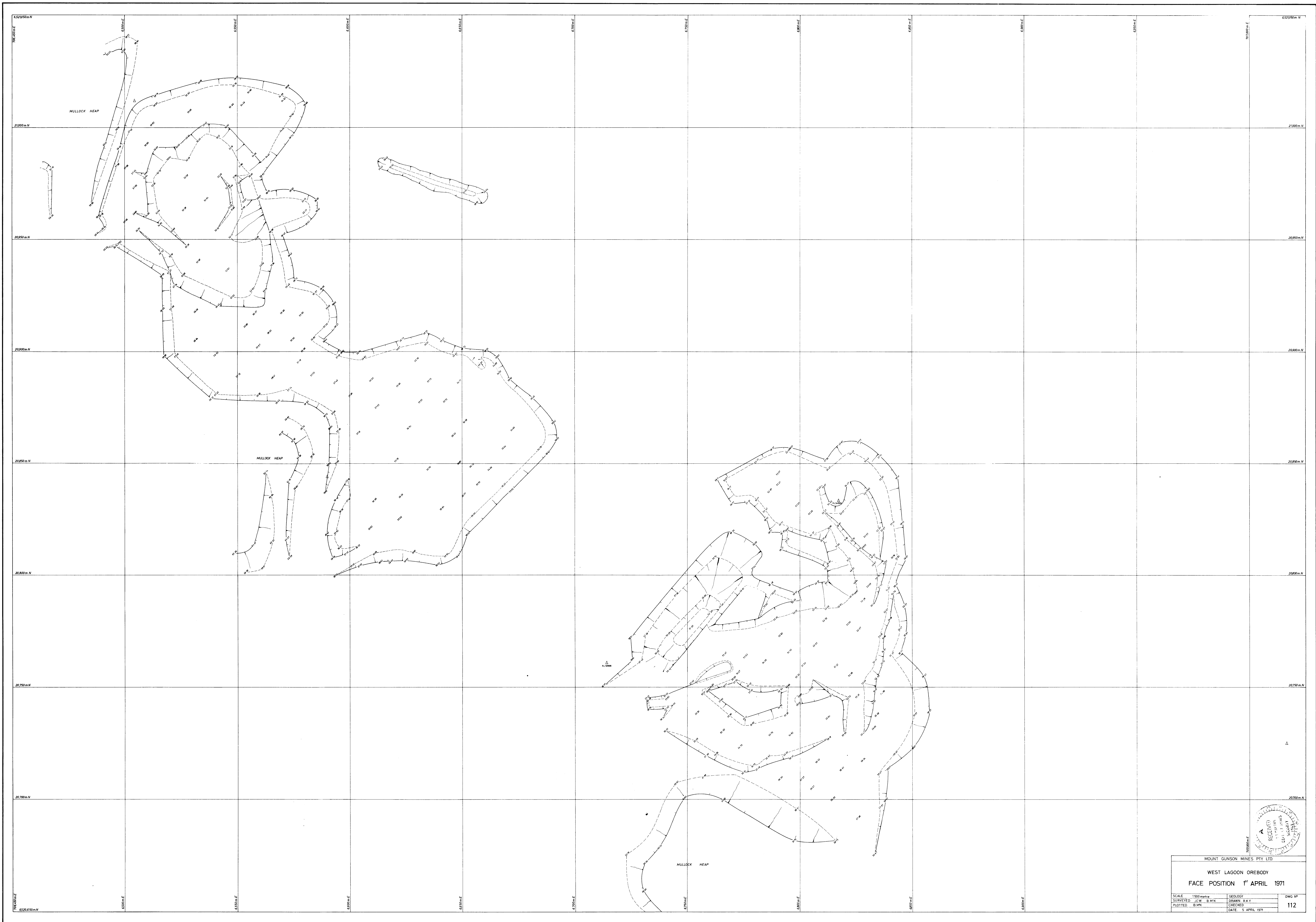




MT. GUNSON ORE CONTROLS AS A  
BASIS FOR FURTHER EXPLORATION

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1. Purpose of This Report

This report is a brief summary of my views on ore controls at the Mt. Gunson Mines and surrounds and incorporates recommendations for exploration.  
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2. Geological Setting

Regionally, copper deposits of Mt. Gunson lie in an inlier of older Pandurra Formation (here termed the "window") projecting through overlying, disconformable Woocalla Dolomite. In turn the "window" is partly capped by the Tent Hill Formation, which lies disconformably over both the lower rock units.

The window of Pandurra Formation, protruding through Woocalla Dolomite, represents a north-trending structural culmination of the gently-folded rock sequence. Within this window, known economic concentrations of copper, as seen in plan, are restricted by the following control:  
^

(1) to the eastern and southern boundaries, close to the Woocalla Dolomite, where (2) this horizon is represented by a dolomite or dolomitic limestone of possible stromatolite reef origin. Elsewhere, the Woocalla Formation does not outcrop, but has been intersected in bores and found to be represented by a black dolomitic shale and not reef-like carbonate (Figs. 1 and 2).

Localisation of copper ores has a further control (i.e. within the window of Pandurra Formation and close to outcropping dolomite) in depressions in the pre-Woocalla erosion surface on the Pandurra Formation.

### 3. Ore Genesis

#### The Source of Copper

It is suggested that <sup>the</sup> ultimate source of the copper metal was <sup>a</sup> dispersed syngenetic sources <sup>^</sup> in the Woocalla Dolomite. This formation contains anomalously high amounts of heavy metals, including copper. On the evidence from a small number of geochemical rock analyses, the formation appears to be especially rich in copper around the areas now being worked and worked in the past, i.e. at the Mt. Gunson Mine area and at Sweet Nell and Monalena, east and south of Woocalla.

#### The Concentration to Copper Ore

The process responsible for the concentration of copper into economic bodies is envisaged as having been a residual upgrading of syngenetic Cu by dissolution under weathering or other <sup>supergene</sup> ~~supergene~~ processes, and removal in solution of dolomite <sup>from</sup> ~~in~~ the rock body <sup>^</sup> exposed by uplift along the Pernatty Culmination. The now upgraded remaining copper rich material is concentrated in depressions in the disconformable, pre-Woocalla surface, with the flanking, unweathered dolomite formation providing a precipitation barrier. The same

process may have been responsible for the formation of manganese bodies of the district.

### Feasibility of the Concentration Mechanism

Copper values of 300 to 500 p.p.m. have been recorded in unmineralised dolomite in areas near copper workings. Lower values were obtained in the area of the main manganese mines.

A content of 300 p.p.m. of Cu in dolomite, is equivalent to a grade of 0.6 pounds of Cu per short ton of rock.

Using a figure of 14 cubic feet of dolomite to the ton, 1 cubic foot of rock would provide  $\frac{0.6}{14}$  pounds of Cu metal.

Three million tons of 1% Cu Mineralization in the Gunson area represents  $60 \times 10^6$  pounds of copper metal. Therefore  $14 \times 10^8$  cubic feet (approx.  $5 \times 10^7$  cubic yards) of dolomite would be required to be dissolved to release the quantity of copper contained in the Gunson deposits if it were originally syngenetic copper.

If an average thickness of 30 feet is taken for the dolomite, the area of rock that would have to be dissolved to release sufficient copper to form the Gunson deposits would be about 1.5 square miles (only about twice the area of the Gunyah Lake), an area forming but a fraction of the weathered and dissected "window" space.

4. Recommendations on Future Exploration

On the basis of the foregoing hypothesis and also on the basis of paleogeographic similarity, the most favourable areas for future copper exploration would be on the exposed and concealed contact of the dolomite facies of the Woocalla Formation along the eastern and southern rims of the "window" of Pandurra Formation.

Further concentration of exploration effort could be effected in two ways:

- a) By systematic rock-chip geochemical sampling of the outcropping dolomite, or shallow sample drilling, to test for haloes of highest copper concentration within the original rock body.
- b) By concentrating sampling in areas of low present relief, which possibly represent erosional depressions in the pre-Tent Hill surface.

Three prime targets are suggested by this hypothesis (Figure 3). These are in order of interest:

- 1) The concealed western edge of the dolomite between the original Main Open Cut workings and the Gunyot workings.
- 2) The Dolomite - Pandurra contact in Lake Dutton, near South Point.

- 3) The Sweet-Nell area, especially beneath Ironstone Lagoon.

Mr. McManus has indicated a fourth target (which also fits the concepts presented here) in an area to the north of the manganese workings, where the Dolomite - Pandurra contact is concealed by Tent Hill Formation.

5. Reference

D. Tonkin, 1968. Report on Lake Macfarlane Region, S.A.,  
Noranda Aust. Limited, Report No. 103.

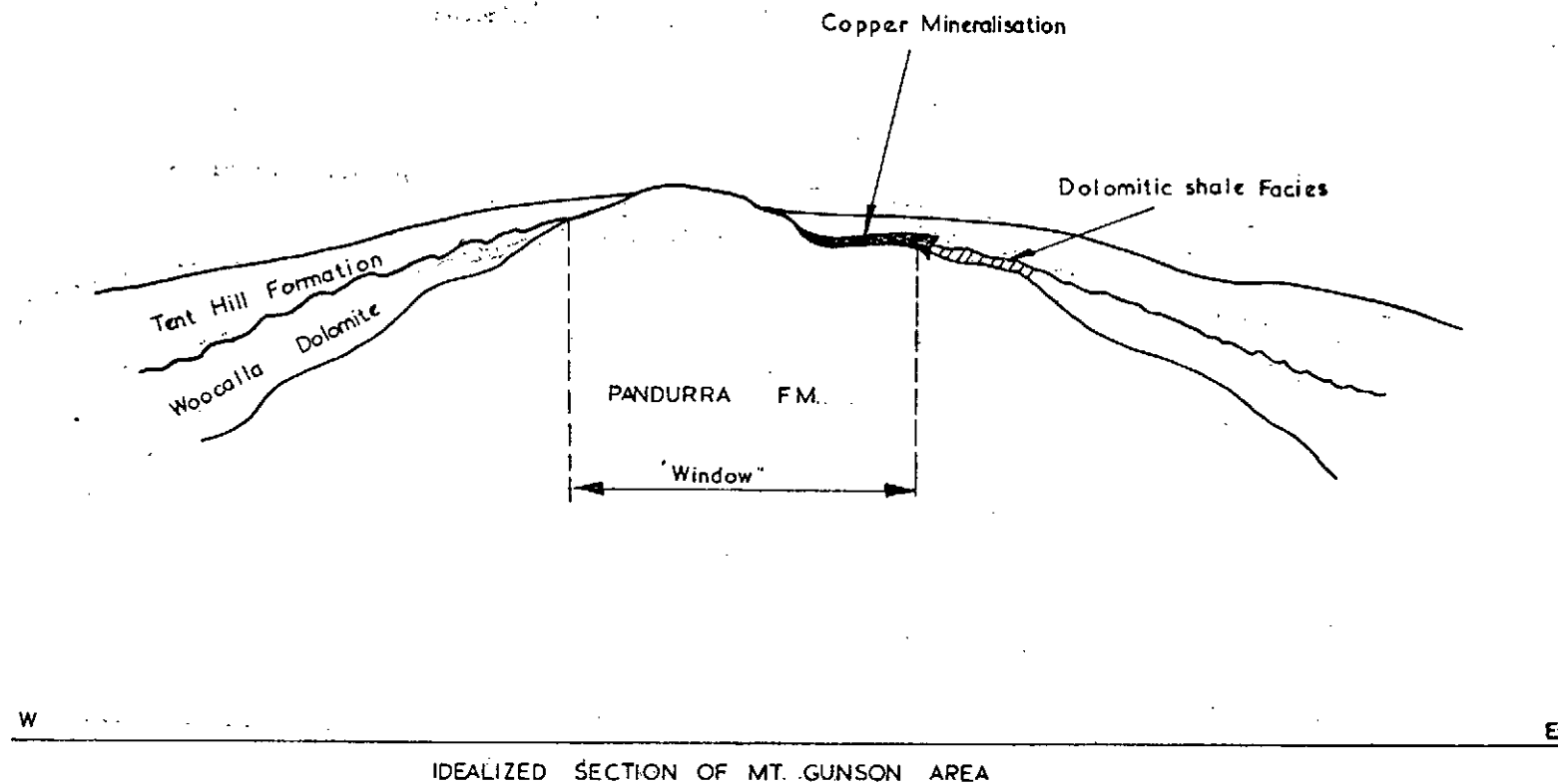


FIGURE 2

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 SECTION  
 SCHEMATIC PLAN OF PRE TENT HILL SURFACE  
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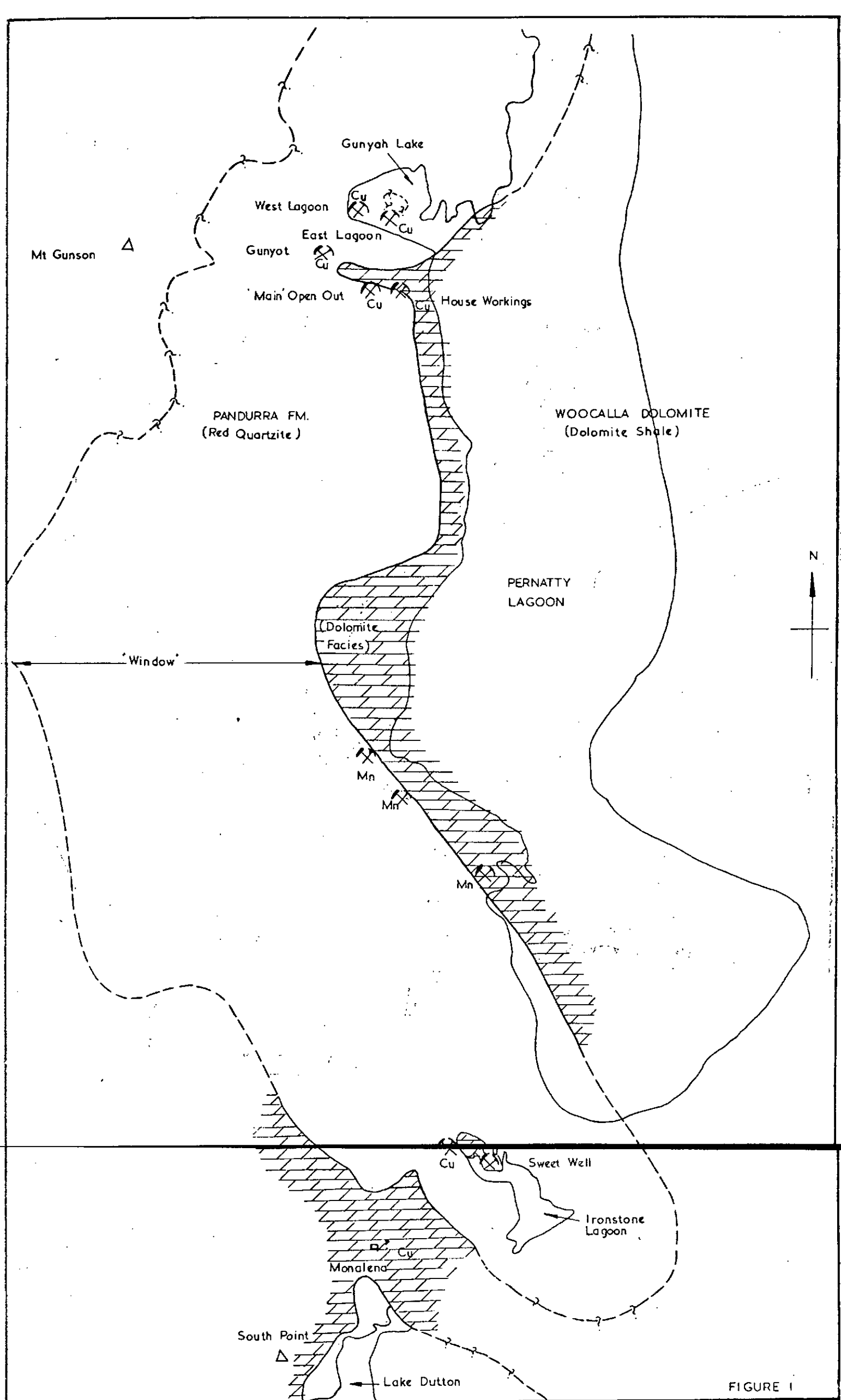


FIGURE 1

PACMINEX PTY. LIMITED  
SCHEMATIC PLAN OF PRE TENT HILL SURFACE

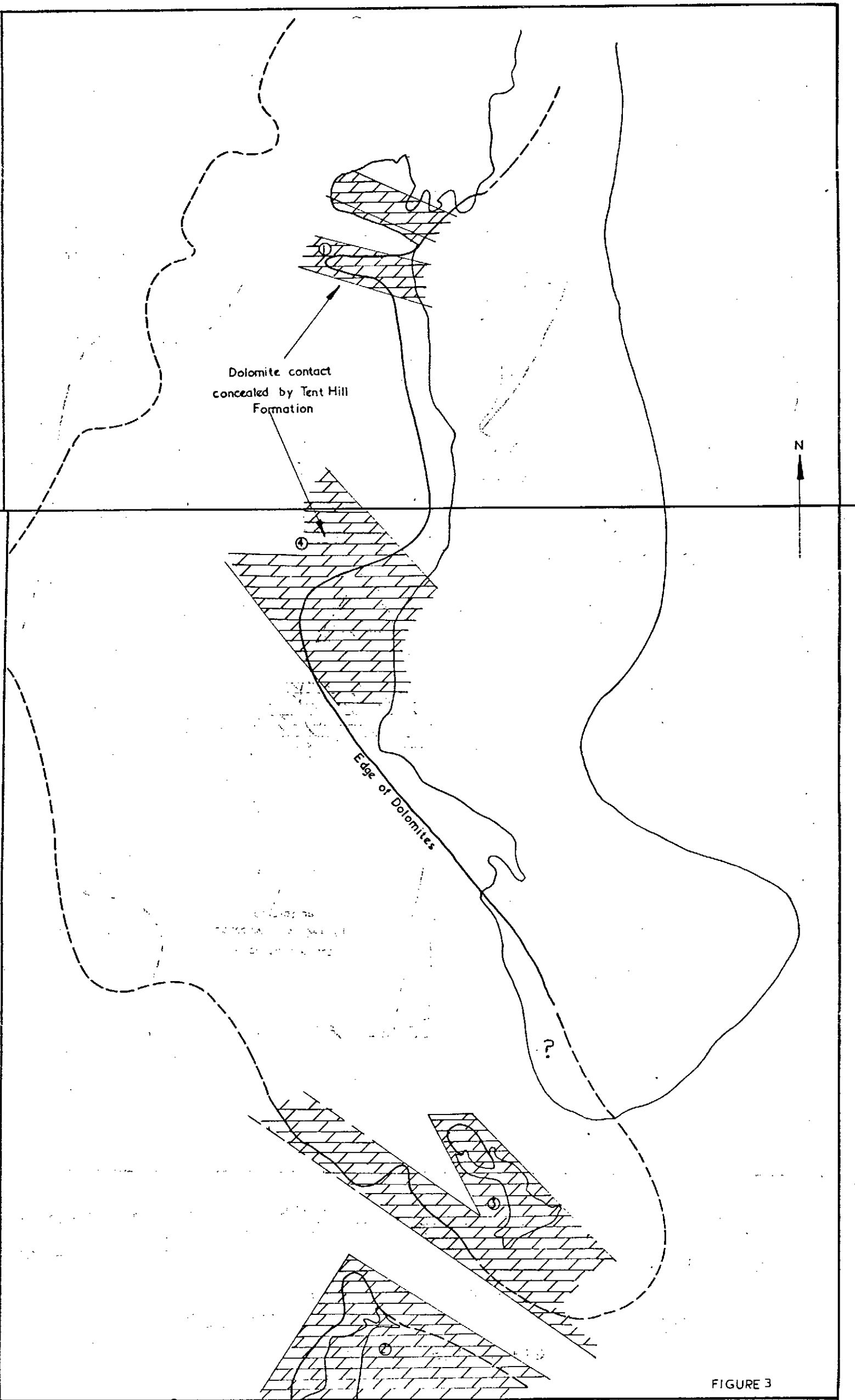
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 PROSPECTIVE AREAS FOR COPPER MINERALISATION  
 MT. GUNSON AREA

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